

Code:

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#include <stdio.h>

#define MAXN 200005

int head[MAXN], to[2 * MAXN], nxt[2 * MAXN], edge_cnt; int matched[MAXN]; int
max_pairs = 0;

void link(int i, int j) { to[++edge_cnt] = j; nxt[edge_cnt] = head[i]; head[i] = edge_cnt; }

void dfs(int p, int i) { for (int e = head[i]; e; e = nxt[e]) { int v = to[e]; if (v != p) { dfs(i, v); } } if
(p != 0 && !matched[p] && !matched[i]) { matched[p] = 1; matched[i] = 1; max_pairs++; } }

int main() { int n; if (scanf("%d", &n) != 1) return 0;

for (int k = 0; k < n - 1; ++k) {
    int a, b;
    scanf("%d %d", &a, &b);
    link(a, b);
    link(b, a);
}

dfs(0, 1);
printf("%d\n", max_pairs);
return 0;

}
```

Input:

```
5
1 2
1 3
3 4
3 5
```

Output:

2